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United States Patent	12408756
Kind Code	B1
Date of Patent	September 09, 2025
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Stacking chair with removable back

Abstract

A stacking chair system comprising a frame with four legs, two being front legs and two being rear legs. The seat is secured to the frame, the seat being upholstered and including a foam cushion supported by flexible support. Two receivers in the frame are at a rear of the frame and between the rear legs of the frame. A backrest comprising a foam cushion is upholstered and has an inner frame supporting the foam cushion, the inner frame including two protrusions extending out of the upholstery wherein the each one of the two protrusions is configured to insert into one of the two receivers. A fastener is provided for each receiver and is configured to secure the protrusion to the corresponding receiver. The backrest is removable upon removal of the fasteners.

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Appl. No.:	**18/325463**
Filed:	**May 30, 2023**

Related U.S. Application Data

us-provisional-application US 63346638 20220527

Publication Classification

Int. Cl.: **A47C3/04** (20060101); **A47C4/02** (20060101); **A47C5/06** (20060101); **A47C5/10** (20060101); **A47C7/02** (20060101); **A47C7/18** (20060101); **A47C7/22** (20060101); **A47C7/24** (20060101); **A47C7/42** (20060101)

U.S. Cl.:

CPC **A47C3/04** (20130101); **A47C4/02** (20130101); **A47C4/028** (20130101); **A47C5/06** (20130101); **A47C5/10** (20130101); **A47C7/0213** (20180801); **A47C7/18** (20130101); **A47C7/22** (20130101); **A47C7/24** (20130101); **A47C7/42** (20130101);

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
62428	12/1866	Koechling	N/A	N/A
146432	12/1873	Close	N/A	N/A
216307	12/1878	Case	N/A	N/A
252533	12/1881	Smith	N/A	N/A
272166	12/1882	Shepler	N/A	N/A
315618	12/1884	Harwood	N/A	N/A
326571	12/1884	Lancaster	N/A	N/A
646290	12/1899	Kratz	N/A	N/A
669089	12/1900	Lambert	N/A	N/A
688878	12/1900	Neuendorff	N/A	N/A
1010557	12/1910	Wilkinson	N/A	N/A
1152480	12/1914	Bouk	N/A	N/A
1374467	12/1920	Park	N/A	N/A
1662378	12/1927	Duke	N/A	N/A
1695101	12/1927	Hoffman	N/A	N/A
1743802	12/1929	Andress	N/A	N/A
1800387	12/1930	Greist	N/A	N/A
1810888	12/1930	Hanson	N/A	N/A
1922582	12/1932	Goodrich	N/A	N/A
1941340	12/1932	Dellert	N/A	N/A
1973178	12/1933	Sass	N/A	N/A
2000172	12/1934	Hanson	N/A	N/A
2044473	12/1935	Geller	N/A	N/A
2070387	12/1936	Vandervoort	N/A	N/A
2098711	12/1936	Rastetter	N/A	N/A
2232205	12/1940	Boardman	N/A	N/A
2290556	12/1941	Hard	N/A	N/A
2304223	12/1941	Westrope	N/A	N/A
2492103	12/1948	Merrill	N/A	N/A
2492111	12/1948	Prosser et al.	N/A	N/A
D156935	12/1949	Carlson	N/A	N/A
2500124	12/1949	Hoven	N/A	N/A
2516728	12/1949	Smith	N/A	N/A
2518381	12/1949	Runkles	N/A	N/A
D165457	12/1950	Hoven et al.	N/A	N/A

2608238	12/1951	Larsen et al.	N/A	N/A
2610335	12/1951	Baptista	N/A	N/A
2610669	12/1951	Bromagem	N/A	N/A
2620858	12/1951	Hoven et al.	N/A	N/A
2621709	12/1951	Bell	N/A	N/A
2652881	12/1952	Rowe	N/A	N/A
2688997	12/1953	Miller	N/A	N/A
2750993	12/1955	McGregor	N/A	N/A
2812120	12/1956	Beall, Jr.	N/A	N/A
2815972	12/1956	Lagervall	N/A	N/A
2828803	12/1957	Howe et al.	N/A	N/A
2842187	12/1957	Hendrickson	N/A	N/A
2853119	12/1957	Balfour	N/A	N/A
2913039	12/1958	Mauser	N/A	N/A
2915223	12/1958	Beall, Jr.	N/A	N/A
2942652	12/1959	Chapman	N/A	N/A
2986201	12/1960	McCartan	N/A	N/A
3087755	12/1962	Boman	N/A	N/A
3098677	12/1962	Williams	N/A	N/A
3102754	12/1962	Junkunc	N/A	N/A
3114575	12/1962	Eames	N/A	N/A
3137527	12/1963	Hoven et al.	N/A	N/A
3170731	12/1964	Caldemeyer et al.	N/A	N/A
3173587	12/1964	Stearns	N/A	N/A
3182377	12/1964	Hoven et al.	N/A	N/A
3194601	12/1964	Hoven et al.	N/A	N/A
3197254	12/1964	Hendrickson	N/A	N/A
3233939	12/1965	Chapman	N/A	N/A
3246927	12/1965	Klassen	N/A	N/A
3264034	12/1965	Lawson	N/A	N/A
3300246	12/1966	Bouche	N/A	N/A
3305044	12/1966	Van Loo et al.	N/A	N/A
3310300	12/1966	Lawson	N/A	N/A
D208468	12/1966	Okiyama	N/A	N/A
3343870	12/1966	Thatcher et al.	N/A	N/A
3369658	12/1967	Hasselmann	N/A	N/A
D210364	12/1967	Lindberg	N/A	N/A
3446530	12/1968	Rowland	N/A	N/A
3479084	12/1968	Van Ryn	N/A	N/A
D216305	12/1968	Gazit	N/A	N/A
3506167	12/1969	Orr	N/A	N/A
3528096	12/1969	Moberg	N/A	N/A
3531552	12/1969	Getz et al.	N/A	N/A
3544163	12/1969	Krein	N/A	N/A
3547488	12/1969	Barnes	N/A	N/A
3550958	12/1969	Krein	N/A	N/A
3556588	12/1970	Monyer et al.	N/A	N/A
3583590	12/1970	Ferraro	N/A	N/A
3589762	12/1970	Henrikson	N/A	N/A
3594037	12/1970	Sherman	N/A	N/A

D221642	12/1970	Bayes	N/A	N/A
D222350	12/1970	Sundberg	N/A	N/A
3641614	12/1971	Newsome	N/A	N/A
3647260	12/1971	Grant et al.	N/A	N/A
3657854	12/1971	Tipton	N/A	N/A
3695687	12/1971	Uyeda	N/A	N/A
3695693	12/1971	Acton et al.	N/A	N/A
3714678	12/1972	Weisz et al.	N/A	N/A
3717289	12/1972	Laurizio	N/A	N/A
3722950	12/1972	Harnick	N/A	N/A
3727975	12/1972	Anderson	N/A	N/A
3744843	12/1972	Barecki et al.	N/A	N/A
D228104	12/1972	Blodee	N/A	N/A
3757984	12/1972	Barton	N/A	N/A
3759572	12/1972	Koepke	N/A	N/A
3762765	12/1972	Piretti	N/A	N/A
3762766	12/1972	Barecki et al.	N/A	N/A
3784249	12/1973	Hendrickson et al.	N/A	N/A
3785600	12/1973	Padovano	N/A	N/A
3796459	12/1973	Weber	N/A	N/A
3813149	12/1973	Lawrence, III et al.	N/A	N/A
3847433	12/1973	Acton et al.	N/A	N/A
3850476	12/1973	Day	N/A	N/A
3857605	12/1973	Fantoni	N/A	N/A
3858766	12/1974	Schiemann	N/A	N/A
3874731	12/1974	Jordan	N/A	N/A
3877674	12/1974	Cerutti	N/A	N/A
3880467	12/1974	Tischler	N/A	N/A
3885766	12/1974	Resch et al.	N/A	N/A
3889999	12/1974	Mackintosh	N/A	N/A
3899207	12/1974	Mueller	N/A	N/A
3901417	12/1974	Schiemann	N/A	N/A
3902754	12/1974	Braeuning	N/A	N/A
3933268	12/1975	Buske	N/A	N/A
3960405	12/1975	DeLong	N/A	N/A
3976326	12/1975	Grumblatt	N/A	N/A
4000586	12/1976	Vance et al.	N/A	N/A
4052101	12/1976	DeLong	N/A	N/A
4054316	12/1976	DeLong	N/A	N/A
4077669	12/1977	Fox	N/A	N/A
4097089	12/1977	Petersen	N/A	N/A
4114947	12/1977	Nelson	N/A	N/A
4126354	12/1977	DeLong et al.	N/A	N/A
4133579	12/1978	Springfield	N/A	N/A
D253034	12/1978	Meyer et al.	N/A	N/A
4179158	12/1978	Flaum et al.	N/A	N/A
4189876	12/1979	Crossman et al.	N/A	N/A
4216994	12/1979	Benoit	N/A	N/A
4240663	12/1979	Locher	N/A	N/A
D258866	12/1980	Houseman et al.	N/A	N/A

4273265	12/1980	Anderson	N/A	N/A
4277101	12/1980	Vogel	N/A	N/A
D260058	12/1980	Ponzellini	N/A	N/A
4297763	12/1980	Lautenschlager	N/A	N/A
4308961	12/1981	Kunce	N/A	N/A
4330898	12/1981	Thompson et al.	N/A	N/A
D266017	12/1981	Kellogg	N/A	N/A
4348053	12/1981	Strassle	N/A	N/A
4453363	12/1983	Koller	N/A	N/A
D274699	12/1983	Epperson	N/A	N/A
4493172	12/1984	Jones	N/A	N/A
4509665	12/1984	Goodall	N/A	N/A
4528786	12/1984	Dinitz et al.	N/A	N/A
4542887	12/1984	Bethell et al.	N/A	N/A
4547092	12/1984	Vetter et al.	N/A	N/A
4575149	12/1985	Forestal et al.	N/A	N/A
4603907	12/1985	Witzke	N/A	N/A
4632457	12/1985	Hofrichter et al.	N/A	N/A
4650100	12/1986	Echazabal, Jr.	N/A	N/A
4691828	12/1986	Slusarczyk et al.	N/A	N/A
4694531	12/1986	Foy	N/A	N/A
4702522	12/1986	Vail et al.	N/A	N/A
4705274	12/1986	Lubeck	N/A	N/A
4723816	12/1987	Selbert et al.	N/A	N/A
4725027	12/1987	Bekanich	N/A	N/A
4726554	12/1987	Sorrell	N/A	N/A
4748913	12/1987	Favaretto et al.	N/A	N/A
4754943	12/1987	Froutzis	N/A	N/A
4761035	12/1987	Urai	N/A	N/A
4766838	12/1987	Johnson	N/A	N/A
4789126	12/1987	Rice et al.	N/A	N/A
4790594	12/1987	Temos	N/A	N/A
4799632	12/1988	Baymak et al.	N/A	N/A
4805793	12/1988	Brandt et al.	N/A	N/A
4828208	12/1988	Peterson et al.	N/A	N/A
4850159	12/1988	Conner	N/A	N/A
4850502	12/1988	Davis	N/A	N/A
4852940	12/1988	Kanigowski	N/A	N/A
4854640	12/1988	Yokoyama	N/A	N/A
4858996	12/1988	Blodee	N/A	N/A
4864690	12/1988	Chen	N/A	N/A
4865377	12/1988	Musser et al.	N/A	N/A
4883319	12/1988	Scott	N/A	N/A
4900085	12/1989	Tobler	N/A	N/A
4955575	12/1989	Moore	N/A	N/A
4960137	12/1989	Pott et al.	N/A	N/A
4984768	12/1990	Kolber et al.	N/A	N/A
4989915	12/1990	Hansal	N/A	N/A
5002337	12/1990	Engel et al.	N/A	N/A
D315867	12/1990	Hestehave et al.	N/A	N/A

5020286	12/1990	Hamar et al.	N/A	N/A
5033792	12/1990	Kanazawa	N/A	N/A
5067772	12/1990	Koa	N/A	N/A
5087096	12/1991	Yamazaki	N/A	N/A
5098156	12/1991	Vogel	N/A	N/A
5108016	12/1991	Waring	N/A	N/A
5110013	12/1991	Clark et al.	N/A	N/A
5121891	12/1991	Goldsmith	N/A	N/A
5123702	12/1991	Caruso	N/A	N/A
5142757	12/1991	Thary	N/A	N/A
D331703	12/1991	Seguin et al.	N/A	N/A
5167336	12/1991	Lajovic	N/A	N/A
D335453	12/1992	Marti	N/A	N/A
5232110	12/1992	Purnell	N/A	N/A
5236165	12/1992	Conway	N/A	N/A
5236247	12/1992	Hewko	N/A	N/A
5246270	12/1992	Vogel	N/A	N/A
5282662	12/1993	Bolsworth et al.	N/A	N/A
5288128	12/1993	Smith	N/A	N/A
5306072	12/1993	Caldwell	N/A	N/A
5308126	12/1993	Weger, Jr. et al.	N/A	N/A
5316375	12/1993	Breen	N/A	N/A
5328231	12/1993	Raymond	N/A	N/A
5329871	12/1993	Gibbs	N/A	N/A
5342110	12/1993	Merrick	N/A	N/A
5375914	12/1993	Donnelly	N/A	N/A
5377882	12/1994	Pham et al.	N/A	N/A
5383644	12/1994	Huse	N/A	N/A
5393126	12/1994	Boulva	N/A	N/A
5400928	12/1994	Resnick	N/A	N/A
5406994	12/1994	Mitchell et al.	N/A	N/A
D358330	12/1994	Kahl	N/A	N/A
5433346	12/1994	Howe	N/A	N/A
5470128	12/1994	Kerkham	N/A	N/A
5472124	12/1994	Martushev	N/A	N/A
D365983	12/1995	Triassi et al.	N/A	N/A
5524963	12/1995	Barile	297/239	A47C 3/04
5529376	12/1995	Jovan et al.	N/A	N/A
5538165	12/1995	Frohn	N/A	N/A
5547247	12/1995	Dixon	N/A	N/A
5549072	12/1995	Maloney	N/A	N/A
5553923	12/1995	Bilezikjian	N/A	N/A
5567016	12/1995	Koprowski	N/A	N/A
5597097	12/1996	Morris	N/A	N/A
5598785	12/1996	Zaguroli, Jr.	N/A	N/A
5601335	12/1996	Woods et al.	N/A	N/A
5609395	12/1996	Burch	N/A	N/A
5632406	12/1996	Robbins, III	N/A	N/A
5636900	12/1996	Wilkie et al.	N/A	N/A
5651908	12/1996	Mansfield	N/A	N/A

5658043	12/1996	Davidson	N/A	N/A
5661945	12/1996	Henriksson et al.	N/A	N/A
5671868	12/1996	Herr	N/A	N/A
5671975	12/1996	Muller	N/A	N/A
5681085	12/1996	Nahoul	N/A	N/A
5681089	12/1996	Lin	N/A	N/A
5683136	12/1996	Baumann et al.	N/A	N/A
5702157	12/1996	Hurite	N/A	N/A
5704691	12/1997	Olson	N/A	N/A
5711355	12/1997	Kowalczyk	N/A	N/A
5733010	12/1997	Lewis et al.	N/A	N/A
D393803	12/1997	Walter	N/A	N/A
5740947	12/1997	Flaig et al.	N/A	N/A
5746358	12/1997	Crosby	N/A	N/A
5762296	12/1997	Gilbert	N/A	N/A
5762396	12/1997	Barile	297/239	A47C 3/04
5765911	12/1997	Sorenson	N/A	N/A
D396967	12/1997	Yamazaki	N/A	N/A
5788332	12/1997	Hettinga	N/A	N/A
5803546	12/1997	Yamazaki	N/A	N/A
5826850	12/1997	Goldsmith	N/A	N/A
5845964	12/1997	Phoon	N/A	N/A
5850949	12/1997	Koerbel et al.	N/A	N/A
5878470	12/1998	Blansett	N/A	N/A
5881431	12/1998	Pieper, II et al.	N/A	N/A
5890761	12/1998	Miller	N/A	N/A
5897035	12/1998	Schlomer	N/A	N/A
5899526	12/1998	LaPointe et al.	N/A	N/A
5899531	12/1998	Koehler	N/A	N/A
5924608	12/1998	Chiu	N/A	N/A
5931528	12/1998	Shields	N/A	N/A
5944387	12/1998	Stumpf	N/A	N/A
5944388	12/1998	Saucier et al.	N/A	N/A
5971466	12/1998	Hashimoto	N/A	N/A
5984417	12/1998	Wang	N/A	N/A
5985188	12/1998	Jennings et al.	N/A	N/A
5988757	12/1998	Vishey et al.	N/A	N/A
6012773	12/1999	Best	N/A	N/A
6019413	12/1999	Scraver et al.	N/A	N/A
6024397	12/1999	Scraver et al.	N/A	N/A
6029858	12/1999	Srokose et al.	N/A	N/A
6045013	12/1999	Yang	N/A	N/A
6050455	12/1999	Soehnlen et al.	N/A	N/A
6056278	12/1999	Bullard et al.	N/A	N/A
6068161	12/1999	Soehnlen et al.	N/A	N/A
D426162	12/1999	Olson	N/A	N/A
6073997	12/1999	Koh	N/A	N/A
6082591	12/1999	Healey	N/A	N/A
6092864	12/1999	Wycech et al.	N/A	N/A
6095603	12/1999	Hock	N/A	N/A

6109696	12/1999	Newhouse	297/452.21	A47C 7/42
6116183	12/1999	Crow et al.	N/A	N/A
6135562	12/1999	Infanti	N/A	N/A
6145912	12/1999	Rice et al.	N/A	N/A
6161262	12/1999	Pfister	N/A	N/A
6174029	12/2000	Swy	297/440.22	A47C 3/04
D438119	12/2000	Meyer et al.	N/A	N/A
6190047	12/2000	Haag et al.	N/A	N/A
6196425	12/2000	Fielding et al.	N/A	N/A
6206469	12/2000	Caruso et al.	N/A	N/A
6220658	12/2000	Lukawski et al.	N/A	N/A
6220661	12/2000	Peterson	N/A	N/A
D441972	12/2000	Mitjans	N/A	N/A
6224149	12/2000	Gevaert	N/A	N/A
6226819	12/2000	Ogawa et al.	N/A	N/A
6230944	12/2000	Castellano et al.	N/A	N/A
D445269	12/2000	Mitjans	N/A	N/A
D445574	12/2000	Foster	N/A	N/A
D446035	12/2000	Haney	N/A	N/A
D447642	12/2000	Kimura	N/A	N/A
6283550	12/2000	Vialatte et al.	N/A	N/A
6286902	12/2000	Yoshimura	N/A	N/A
6293621	12/2000	Marsh et al.	N/A	N/A
6296222	12/2000	Whittaker	N/A	N/A
6296315	12/2000	Jensen	N/A	N/A
D450245	12/2000	Schultz	N/A	N/A
6314614	12/2000	Kuehl	N/A	N/A
6345863	12/2001	Laws et al.	N/A	N/A
6349449	12/2001	Kuehl	N/A	N/A
6352307	12/2001	Engman	N/A	N/A
6360924	12/2001	Franzen	N/A	N/A
6382475	12/2001	Weikinger	N/A	N/A
6382726	12/2001	Bullesbach et al.	N/A	N/A
6393624	12/2001	Iwashita	N/A	N/A
D458140	12/2001	Hutchins	N/A	N/A
6409267	12/2001	Pietrzak	N/A	N/A
6425153	12/2001	Reswick	N/A	N/A
6427287	12/2001	Brueckner et al.	N/A	N/A
6428100	12/2001	Kain et al.	N/A	N/A
6450579	12/2001	Nylander et al.	N/A	N/A
D465664	12/2001	Stannard	N/A	N/A
6474744	12/2001	Taylor et al.	N/A	N/A
6477738	12/2001	Kremer et al.	N/A	N/A
6478381	12/2001	Cramb, III et al.	N/A	N/A
6491346	12/2001	Gupta et al.	N/A	N/A
6494344	12/2001	Kressel, Sr.	N/A	N/A
6499158	12/2001	Easterling	N/A	N/A
6523900	12/2002	Conner et al.	N/A	N/A
6554353	12/2002	Yu	N/A	N/A
6565157	12/2002	Barile, Jr. et al.	N/A	N/A

6578922	12/2002	Khedira et al.	N/A	N/A
6582020	12/2002	Tenenboym et al.	N/A	N/A
6588612	12/2002	Dorn et al.	N/A	N/A
D479468	12/2002	Dorn et al.	N/A	N/A
D479657	12/2002	Calatrava Valls	N/A	N/A
6612652	12/2002	Tenenboym et al.	N/A	N/A
6669282	12/2002	Piretti	N/A	N/A
D485170	12/2003	Hay et al.	N/A	N/A
6683394	12/2003	Gevaert	N/A	N/A
6698834	12/2003	Olarte	N/A	N/A
6729685	12/2003	Ebalobor	N/A	N/A
6746086	12/2003	Foster	N/A	N/A
6786548	12/2003	Pearce et al.	N/A	N/A
6786549	12/2003	Olarte	N/A	N/A
6789549	12/2003	Johnson	N/A	N/A
6793281	12/2003	Duerr et al.	N/A	N/A
6805394	12/2003	Kettler et al.	N/A	N/A
D498077	12/2003	Tonizzo	N/A	N/A
6839950	12/2004	Guillot	N/A	N/A
6845885	12/2004	Morgenroth	N/A	N/A
6857699	12/2004	Ashton et al.	N/A	N/A
6874850	12/2004	Berkowicz	N/A	N/A
D506137	12/2004	Durivage	N/A	N/A
6907703	12/2004	Gonzalez	N/A	N/A
6932228	12/2004	Darr et al.	N/A	N/A
D509144	12/2004	Snyder	N/A	N/A
6942295	12/2004	Lopez	N/A	N/A
6964345	12/2004	Wetherell, Jr. et al.	N/A	N/A
6968983	12/2004	Laible	N/A	N/A
D514940	12/2005	Darr et al.	N/A	N/A
7000794	12/2005	Soehnlen et al.	N/A	N/A
7000989	12/2005	Fisher	N/A	N/A
D517413	12/2005	Murphy	N/A	N/A
D519378	12/2005	Abadinsky	N/A	N/A
7032272	12/2005	Haenlein	N/A	N/A
D520367	12/2005	Morris	N/A	N/A
D520765	12/2005	Piano	N/A	N/A
7063381	12/2005	Scahill et al.	N/A	N/A
7073858	12/2005	Fisher et al.	N/A	N/A
D528425	12/2005	Van Dorin et al.	N/A	N/A
7100983	12/2005	Gant	N/A	N/A
7108447	12/2005	Akkala et al.	N/A	N/A
D530936	12/2005	Gehry	N/A	N/A
7134728	12/2005	Buhrman	N/A	N/A
7150498	12/2005	Svartvatn	N/A	N/A
7182214	12/2006	Darr et al.	N/A	N/A
D537648	12/2006	Moschini	N/A	N/A
7204553	12/2006	Olarte	N/A	N/A
7210735	12/2006	Lang	N/A	N/A
7225899	12/2006	Molnar et al.	N/A	N/A

D549983	12/2006	Olarte	N/A	N/A
D549984	12/2006	Olarte	N/A	N/A
D556058	12/2006	Bowers et al.	N/A	N/A
7303235	12/2006	Fongers	N/A	N/A
D558604	12/2007	Dohm et al.	N/A	N/A
D559108	12/2007	Regas	N/A	N/A
D560921	12/2007	Olarte	N/A	N/A
7331490	12/2007	Yamana	N/A	N/A
7370910	12/2007	Piretti	N/A	N/A
7384102	12/2007	Chen et al.	N/A	N/A
D573473	12/2007	Kruparova	N/A	N/A
D577589	12/2007	Brieussel et al.	N/A	N/A
7467453	12/2007	Olarte	N/A	N/A
7469966	12/2007	Vallee	N/A	N/A
7478876	12/2008	Olarte	N/A	N/A
7490392	12/2008	Peterson	N/A	N/A
D589807	12/2008	Gundrum et al.	N/A	N/A
D589809	12/2008	Gundrum et al.	N/A	N/A
7513394	12/2008	Bone	N/A	N/A
D594669	12/2008	Asano	N/A	N/A
7562937	12/2008	Baumann	N/A	N/A
7594696	12/2008	Girard	N/A	N/A
7661549	12/2009	Rae	N/A	N/A
7690732	12/2009	Olarte	N/A	N/A
7695061	12/2009	Olarte	N/A	N/A
D616658	12/2009	Niermeyer	N/A	N/A
7774880	12/2009	Botts	N/A	N/A
D625774	12/2009	Nirenberg	N/A	N/A
7814636	12/2009	Olarte	N/A	N/A
7828380	12/2009	Olarte	N/A	N/A
7866689	12/2010	Saberan	N/A	N/A
7882795	12/2010	Snyder	N/A	N/A
7901004	12/2010	Figueras Mitjans	N/A	N/A
7905546	12/2010	Conner	N/A	N/A
7950739	12/2010	King	N/A	N/A
7959044	12/2010	Christian et al.	N/A	N/A
7971935	12/2010	Saez et al.	N/A	N/A
8020254	12/2010	Lin	N/A	N/A
8075055	12/2010	Olarte	N/A	N/A
8091744	12/2011	Zwahlen	N/A	N/A
8091945	12/2011	Hancock et al.	N/A	N/A
8104838	12/2011	Tsai	N/A	N/A
8109566	12/2011	Koh	N/A	N/A
8157132	12/2011	Johnson et al.	N/A	N/A
8201588	12/2011	Bonner	N/A	N/A
8201699	12/2011	Zummo et al.	N/A	N/A
8245891	12/2011	Eriksen	N/A	N/A
D672657	12/2011	Hall	N/A	N/A
D672980	12/2011	Mitjans	N/A	N/A
8414072	12/2012	Phillips	N/A	N/A

8490335	12/2012	LaForest	N/A	N/A
8491053	12/2012	Stringer	N/A	N/A
D687642	12/2012	Hsuan-Chin	N/A	N/A
D687921	12/2012	Jung et al.	N/A	N/A
D687922	12/2012	Jung et al.	N/A	N/A
D689297	12/2012	Su	N/A	N/A
D690203	12/2012	Hall et al.	N/A	N/A
D690952	12/2012	Becker	N/A	N/A
D693426	12/2012	Mehra et al.	N/A	N/A
8573704	12/2012	Peters et al.	N/A	N/A
8578974	12/2012	Bonner	N/A	N/A
D696956	12/2013	Jung et al.	N/A	N/A
8640930	12/2013	Nunez et al.	N/A	N/A
8662322	12/2013	Magnusson et al.	N/A	N/A
8662359	12/2013	Hickey	N/A	N/A
D703458	12/2013	Nakamura et al.	N/A	N/A
8721003	12/2013	Waite et al.	N/A	N/A
8727187	12/2013	Magley	N/A	N/A
8727445	12/2013	De Maina	N/A	N/A
8763826	12/2013	Smith et al.	N/A	N/A
8813657	12/2013	Winter et al.	N/A	N/A
8820827	12/2013	Olarte	N/A	N/A
8820836	12/2013	Stewart et al.	N/A	N/A
8827369	12/2013	Izawa et al.	N/A	N/A
8833617	12/2013	Compton et al.	N/A	N/A
8910835	12/2013	Ouderkirk	N/A	N/A
8960454	12/2014	Magnusson et al.	N/A	N/A
8967714	12/2014	Machael et al.	N/A	N/A
8968041	12/2014	Rubbo	N/A	N/A
8973990	12/2014	Krupiczewicz et al.	N/A	N/A
9033414	12/2014	Olarte	N/A	N/A
9060614	12/2014	Gibilterra	N/A	N/A
D734621	12/2014	Su	N/A	N/A
9096357	12/2014	Brausen	N/A	N/A
9113715	12/2014	Olarte	N/A	N/A
9295334	12/2015	Olarte	N/A	N/A
9380870	12/2015	Babick	N/A	N/A
9399876	12/2015	Magnus	N/A	N/A
D764295	12/2015	Gieske et al.	N/A	N/A
9420890	12/2015	Lee	N/A	N/A
9565949	12/2016	Peterson	N/A	N/A
D781107	12/2016	Welsch et al.	N/A	N/A
9630539	12/2016	Yokoyama et al.	N/A	N/A
D786575	12/2016	Olarte	N/A	N/A
9642464	12/2016	Magnus	N/A	N/A
9669972	12/2016	Stratton	N/A	N/A
9693630	12/2016	Jacobs et al.	N/A	N/A
9693631	12/2016	Jacobs et al.	N/A	N/A
D804313	12/2016	Reches et al.	N/A	N/A
D808676	12/2017	Bellissimo	N/A	N/A

9873354	12/2017	Poulos et al.	N/A	N/A
D810573	12/2017	Vazin	N/A	N/A
9926104	12/2017	Olarte	N/A	N/A
D821880	12/2017	Olarte	N/A	N/A
10010178	12/2017	Maughan	N/A	N/A
10273056	12/2018	Hoffman et al.	N/A	N/A
D847527	12/2018	Borras Pons	N/A	N/A
10315800	12/2018	Olarte	N/A	N/A
10349749	12/2018	Su	N/A	N/A
10485348	12/2018	Olarte	N/A	N/A
10517400	12/2018	Olarte	N/A	A47C 3/045
10518933	12/2018	Dorn	N/A	N/A
10555610	12/2019	Jacobs et al.	N/A	N/A
D876852	12/2019	Bellissimo	N/A	N/A
10588414	12/2019	Olarte	N/A	N/A
10647471	12/2019	Olarte	N/A	N/A
10674825	12/2019	Olarte	N/A	N/A
10681983	12/2019	Olarte	N/A	N/A
10829277	12/2019	Olarte	N/A	N/A
10835045	12/2019	Olarte	N/A	N/A
D906118	12/2019	Olarte	N/A	N/A
10881210	12/2020	Olarte	N/A	N/A
10944890	12/2020	Tsuchiya et al.	N/A	N/A
10994890	12/2020	Olarte	N/A	N/A
11008108	12/2020	Gilbert	N/A	N/A
11026515	12/2020	Olarte	N/A	N/A
11071389	12/2020	Olarte	N/A	N/A
11224292	12/2021	Hashimoto et al.	N/A	N/A
11457744	12/2021	Olarte	N/A	N/A
11464337	12/2021	Olarte	N/A	N/A
11470969	12/2021	Olarte	N/A	N/A
11505367	12/2021	Olarte	N/A	N/A
D979272	12/2022	Olarte	N/A	N/A
11684160	12/2022	Olarte	N/A	N/A
11690458	12/2022	Policicchio	297/440.14	A47C 13/005
11712118	12/2022	Olarte	N/A	N/A
11737568	12/2022	Olarte	N/A	N/A
11793315	12/2022	Olarte	N/A	N/A
11800932	12/2022	Olarte	N/A	N/A
D1019161	12/2023	Olarte	N/A	N/A
12011099	12/2023	Olarte	N/A	N/A
12064035	12/2023	Olarte	N/A	N/A
12150558	12/2023	Olarte	N/A	N/A
2001/0023510	12/2000	Masubuchi	N/A	N/A
2001/0054628	12/2000	Harbaugh et al.	N/A	N/A
2002/0038970	12/2001	Leng et al.	N/A	N/A
2002/0053822	12/2001	Ware	297/239	A47C 7/445

2002/0117885	12/2001	Barile, Jr.	297/452.52	A47C 7/282
2002/0166857	12/2001	Schwaikert	N/A	N/A
2002/0167789	12/2001	Novin et al.	N/A	N/A
2003/0047976	12/2002	Hannon et al.	N/A	N/A
2003/0062759	12/2002	Gupta et al.	N/A	N/A
2003/0085607	12/2002	Jones et al.	N/A	N/A
2003/0090136	12/2002	Olarte	N/A	N/A
2003/0102703	12/2002	Tenenboym et al.	N/A	N/A
2003/0132184	12/2002	Dorn et al.	N/A	N/A
2004/0041455	12/2003	Tenbroeck et al.	N/A	N/A
2004/0084943	12/2003	Fisher	N/A	N/A
2004/0099672	12/2003	Perlman	N/A	N/A
2004/0100134	12/2003	Plant et al.	N/A	N/A
2004/0178222	12/2003	Clausen et al.	N/A	N/A
2004/0217635	12/2003	Barile, Sr.	297/448.1	A47C 3/04
2005/0017554	12/2004	Mizelle et al.	N/A	N/A
2005/0017558	12/2004	Fewchuk	N/A	N/A
2005/0051575	12/2004	Durivage	N/A	N/A
2005/0092780	12/2004	Yamana	N/A	N/A
2005/0099052	12/2004	Bertolini et al.	N/A	N/A
2005/0127740	12/2004	Dowty	N/A	N/A
2005/0140186	12/2004	Piretti	N/A	N/A
2005/0146180	12/2004	Fisher et al.	N/A	N/A
2005/0168032	12/2004	Olarte	N/A	N/A
2005/0173954	12/2004	Weber et al.	N/A	N/A
2005/0194823	12/2004	Perry et al.	N/A	N/A
2005/0194829	12/2004	Aerts	N/A	N/A
2005/0242651	12/2004	Peitz et al.	N/A	N/A
2005/0264060	12/2004	Cesaroni et al.	N/A	N/A
2006/0037930	12/2005	Darr et al.	N/A	N/A
2006/0080817	12/2005	Klinker	N/A	N/A
2006/0081662	12/2005	Miura	N/A	N/A
2006/0202536	12/2005	Luchetti et al.	N/A	N/A
2006/0238004	12/2005	Conner et al.	N/A	N/A
2006/0244301	12/2005	Jeffries	N/A	N/A
2006/0255642	12/2005	Epaud et al.	N/A	N/A
2007/0017886	12/2006	Kao	N/A	N/A
2007/0018495	12/2006	Xiang	297/452.59	A47C 31/02
2007/0029858	12/2006	Grable et al.	N/A	N/A
2007/0040090	12/2006	Fay	N/A	N/A
2007/0063568	12/2006	Figueras Mitjans	N/A	N/A
2007/0120402	12/2006	Bonzini	N/A	N/A
2007/0138849	12/2006	Olarte	N/A	N/A
2007/0170759	12/2006	Nolan et al.	N/A	N/A
2007/0210638	12/2006	Adragna et al.	N/A	N/A
2007/0216205	12/2006	Davis	N/A	N/A
2008/0036266	12/2007	Battey et al.	N/A	N/A
2008/0084104	12/2007	VanHorn	N/A	N/A

2008/0105173	12/2007	Weber	N/A	N/A
2008/0203790	12/2007	Olarte	N/A	N/A
2008/0284215	12/2007	Stauber	N/A	N/A
2009/0039078	12/2008	Sanfilippo et al.	N/A	N/A
2009/0045226	12/2008	Munlin	N/A	N/A
2009/0051207	12/2008	Behrens et al.	N/A	N/A
2009/0121534	12/2008	Brauning	N/A	N/A
2009/0139054	12/2008	Burnley	N/A	N/A
2009/0255893	12/2008	Zummo et al.	N/A	N/A
2009/0273211	12/2008	Hancock et al.	N/A	N/A
2009/0284057	12/2008	Kulish	297/239	A47C 3/04
2010/0171345	12/2009	Powell et al.	N/A	N/A
2010/0176646	12/2009	Rowland et al.	N/A	N/A
2010/0180399	12/2009	Patzer et al.	N/A	N/A
2010/0194160	12/2009	Machael et al.	N/A	N/A
2010/0201165	12/2009	Dankovich	N/A	N/A
2010/0289306	12/2009	Saul	297/239	A47C 3/04
2011/0029123	12/2010	Olarte	N/A	N/A
2011/0108691	12/2010	Alves et al.	N/A	N/A
2011/0115269	12/2010	Roeder et al.	N/A	N/A
2011/0198902	12/2010	Renbarger et al.	N/A	N/A
2011/0233236	12/2010	Brown et al.	N/A	N/A
2012/0126596	12/2011	Olarte	297/239	A47C 3/04
2012/0193318	12/2011	Meager	N/A	N/A
2012/0228338	12/2011	Magley, II	N/A	N/A
2012/0228918	12/2011	Takahashi et al.	N/A	N/A
2012/0247313	12/2011	Peters	N/A	N/A
2012/0248837	12/2011	Peters	N/A	N/A
2012/0261971	12/2011	Olarte	N/A	N/A
2012/0280001	12/2011	Leadbeater	N/A	N/A
2012/0325718	12/2011	Larsen	N/A	N/A
2013/0057036	12/2012	Olarte	N/A	N/A
2013/0082500	12/2012	Line et al.	N/A	N/A
2013/0255797	12/2012	Coulon et al.	N/A	N/A
2013/0299522	12/2012	Magley, II	N/A	N/A
2014/0021222	12/2013	Forbis et al.	N/A	N/A
2014/0144948	12/2013	Brausen	N/A	N/A
2014/0165331	12/2013	Kang et al.	N/A	N/A
2014/0225404	12/2013	Ramirez Magaña	N/A	N/A
2014/0232153	12/2013	Bell et al.	N/A	N/A
2014/0239678	12/2013	Todd	N/A	N/A
2014/0252836	12/2013	Olarte	N/A	N/A
2014/0283337	12/2013	Triebold et al.	N/A	N/A
2014/0325920	12/2013	Ugolini et al.	N/A	N/A
2014/0332568	12/2013	Stratton	N/A	N/A
2015/0015051	12/2014	Huang	297/445.1	A47C 3/04
2015/0021964	12/2014	Jacobs et al.	N/A	N/A
2015/0028641	12/2014	Stent et al.	N/A	N/A
2015/0054322	12/2014	Jacobs et al.	N/A	N/A
2015/0130236	12/2014	Rogers	N/A	N/A

2015/0150380	12/2014	Oh et al.	N/A	N/A
2015/0150381	12/2014	Lee	N/A	N/A
2015/0158213	12/2014	Mogi et al.	N/A	N/A
2015/0282622	12/2014	Kumazawa	N/A	N/A
2015/0296986	12/2014	Rogers	N/A	N/A
2015/0374130	12/2014	Jacobs et al.	N/A	N/A
2016/0255960	12/2015	Piretti	N/A	A47C 7/002
2016/0288910	12/2015	Udriste et al.	N/A	N/A
2016/0316916	12/2015	Pectol	N/A	A47C 7/445
2016/0331138	12/2015	Olarte	N/A	N/A
2016/0331143	12/2015	Johnson et al.	N/A	N/A
2017/0029168	12/2016	Olarte	N/A	N/A
2017/0057130	12/2016	Sameshima et al.	N/A	N/A
2017/0079440	12/2016	Todd	N/A	N/A
2017/0127836	12/2016	Maughan	N/A	A47C 3/04
2017/0164746	12/2016	Phillips	N/A	N/A
2017/0297766	12/2016	Olarte	N/A	N/A
2017/0320415	12/2016	Mueller et al.	N/A	N/A
2017/0327281	12/2016	Cross	N/A	N/A
2018/0098634	12/2017	Smith et al.	N/A	N/A
2018/0177300	12/2017	Jacobs et al.	N/A	N/A
2019/0176669	12/2018	Noro et al.	N/A	N/A
2019/0281973	12/2018	Coates et al.	N/A	N/A
2019/0329673	12/2018	Popescu	N/A	N/A
2019/0374035	12/2018	Olarte	N/A	N/A
2020/0154892	12/2019	Olarte	N/A	N/A
2020/0189739	12/2019	Schultz et al.	N/A	N/A
2020/0216178	12/2019	Lin et al.	N/A	N/A
2021/0052077	12/2020	Smith et al.	N/A	N/A
2021/0219728	12/2020	Plant et al.	N/A	N/A
2021/0321776	12/2020	Plant et al.	N/A	N/A
2021/0393035	12/2020	Arnold	N/A	A47C 4/03
2022/0047082	12/2021	Andersson	N/A	F16B 12/44
2022/0361677	12/2021	Olarte	N/A	N/A
2023/0047482	12/2022	Olarte	N/A	N/A
2023/0312107	12/2022	Valdes de la Garza	N/A	N/A
2023/0320489	12/2022	Olarte	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	**Application Date**	**Country**	**CPC**
2008219128	12/2007	AU	N/A
2019204076	12/2019	AU	N/A
2019204079	12/2019	AU	N/A
2465896	12/2002	CA	N/A
2465896	12/2002	CA	N/A
2678854	12/2007	CA	N/A

2678854	12/2007	CA	N/A
2847538	12/2012	CA	N/A
2847538	12/2012	CA	N/A
2904885	12/2013	CA	N/A
2904885	12/2013	CA	N/A
3045820	12/2018	CA	N/A
3045820	12/2018	CA	N/A
3045828	12/2018	CA	N/A
3045828	12/2018	CA	N/A
3116366	12/2019	CA	N/A
3116366	12/2019	CA	N/A
3152059	12/2021	CA	N/A
3152059	12/2021	CA	N/A
202908255	12/2012	CN	N/A
208610259	12/2018	CN	N/A
1821874	12/1959	DE	N/A
19530156	12/1996	DE	N/A
102007024550	12/2007	DE	N/A
0188002	12/1985	EP	N/A
0241628	12/1986	EP	A47C 4/02
2114199	12/2011	EP	N/A
2114199	12/2011	EP	N/A
2167714	12/1972	FR	N/A
180054	12/1921	GB	N/A
454332	12/1935	GB	N/A
493316	12/1937	GB	N/A
637911	12/1949	GB	N/A
1133972	12/1967	GB	N/A
1464502	12/1976	GB	N/A
2098471	12/1981	GB	N/A
2274391	12/1993	GB	N/A
2343443	12/1999	GB	N/A
01175564	12/1988	JP	N/A
06198087	12/1993	JP	N/A
H1191754	12/1998	JP	N/A
2005527438	12/2004	JP	N/A
2012249948	12/2011	JP	N/A
200341044	12/2003	KR	N/A
WO9608985	12/1994	WO	N/A
WO0016661	12/1999	WO	N/A
2003039297	12/2002	WO	N/A
WO2003039297	12/2002	WO	N/A
2005069799	12/2004	WO	N/A
WO2005069799	12/2004	WO	N/A
WO2007034007	12/2006	WO	N/A
2008103323	12/2007	WO	N/A
WO2008103323	12/2007	WO	N/A
WO2011022585	12/2010	WO	N/A
WO2012054799	12/2011	WO	N/A
2012142394	12/2011	WO	N/A

WO2012142394	12/2011	WO	N/A
2013036514	12/2012	WO	N/A
WO2013036514	12/2012	WO	N/A
WO2013172529	12/2012	WO	N/A
2014164246	12/2013	WO	N/A
WO2014164246	12/2013	WO	N/A
2016183299	12/2015	WO	N/A
WO2016183299	12/2015	WO	N/A
WO2017155562	12/2016	WO	N/A
WO2019065589	12/2018	WO	N/A
2020102690	12/2019	WO	N/A
WO2020102690	12/2019	WO	N/A

OTHER PUBLICATIONS

Canadian Office Action, PCT US2019061755, Jul. 27, 2022. cited by applicant
IP Australia, Australian Government Patent Examination Report No. 1 Application No. 2012304735, Jun. 18, 2015. cited by applicant

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Background/Summary

FIELD OF THE INVENTION
(1) The following relates to chairs and more particularly moveable chairs which can be used when needed and taken out of the way when not.
BACKGROUND OF THE INVENTION
(2) Stacking chairs are used in event spaces that require flexibility. These chairs will be placed on carts when not in use and put into storage. When needed for an event, the chairs can be arranged in rows for e.g. a conference/convention or church and then moved out of the way when a more open space plan is needed. As a result of these chairs being moved much more frequently than chairs which are bolted to the floor (fixed chairs), the upholstery on these moveable chairs can be come damaged and dirty more frequently.
(3) Furthermore, some known designs of moveable chairs can be very expensive to transport in large quantities due to their construction.
SUMMARY OF THE INVENTION
(4) It is therefore an object of the present invention to provide a comfortable stacking chair which allows for easy replacement of upholstery.
(5) It is a further object of the present invention to provide a sturdy an economical stacking chair which can be easily broken down for shipping to conserve space.
(6) These and other objects are achieved by providing a stacking chair whose back can be removed and seat removed to allow for both easy shipping and easy removal and replacement of the upholstery.
(7) In certain aspects the stacking chair system is provided comprising a frame with four legs, two being front legs and two being rear legs. The seat is secured to the frame in a fixed position, the seat being upholstered and including a foam cushion supported by flexible support. Two receivers are provided in the frame at a rear of the frame. The backrest includes a foam cushion which is

upholstered and an inner frame supporting the foam cushion, the inner frame including two protrusions extending out of the upholstery wherein the each one of the two protrusions is configured to insert into one of the two receivers. A fastener is provided for each receiver and is configured to secure the protrusion to the corresponding receiver, wherein the backrest is removable upon removal of the fasteners.
(8) In certain aspects the frame includes a support bar extending on either side of the seat and connected to a front curved section and a rear curved section, the front curved section connected to one of the front legs and the rear curved section connected to one of the rear legs, each of the four legs extending outwards and downwards with respect to the seat and the four legs in a fixed arrangement with respect to the seat. In other aspects the two front legs are arranged at a more vertical angle than the rear legs. In other aspects the seat includes rigid supports arranged around a peripheral edge thereof to define a rigid frame and wherein the flexible support is secured to the rigid frame and the rigid frame is removably secured to a side section of the frame. In still other aspects the foam cushion is arranged on top of the rigid frame.
(9) In certain aspects the frame includes a rear cross bar and the each of two receivers are arranged as a tube extending from the cross bar, the two receivers spaced apart at a distance which is at least 50% or more a width of the seat, but less than a spacing between the rear legs. In still other aspects the seat includes a seat frame and the flexible support is secured to the seat frame, the seat frame includes a curved portion such that the seat includes a lower curved section which matches a curve of the frame adjacent the front legs.
(10) In yet other aspects the seat frame is secured to the frame such that the seat frame rests on top of the frame with upholstery between the seat and the seat frame and fasteners passing through the frame and securing to the seat frame. In yet other aspects the receiver defines an opening with four flat sides. In still other aspects the protrusion includes a circular cross section. In other aspects the protrusion includes a boss defining a flat section and having a threaded feature adjacent thereto for receiving the fasteners. In other aspects the frame includes a rear cross bar and the receivers are secured to the rear cross bar at an attachment point which is lower than a lower surface of a flat section of the frame which is perpendicular to the rear cross bar.
(11) In certain aspects a stacking chair system is provided including a frame with four legs, two being front legs and two being rear legs. A seat is secured to the frame in a fixed position, the seat including a seat frame which defines a perimeter with an interior opening and the seat frame includes at least a flat portion and a curved portion, the curved portion having a curve which matches a curve of the frame ad or adjacent the front legs and the flat portion configured to rest on top of flat portions of the frame, the seat frame contained in upholstery and the seat including a foam cushion positioned above the seat frame wherein the foam cushion is supported by a flexible support, the flexible support connected to the seat frame and extending across the interior opening. In other aspects seat fasteners pass through holes in the flat portions of the frame and through the upholstery to secure to threaded members of the seat frame. In still other aspects at least part of the upholstery is positioned between the seat frame and the flat portions of the frame with the upholstery in contact with the flat portions of the frame. In yet other aspects two receivers are provided in the frame at a rear of the frame and between the rear legs of the frame and the backrest includes a foam cushion which is upholstered and an inner frame supporting the foam cushion. The inner frame includes two protrusions extending out of the upholstery wherein the each one of the two protrusions is configured to insert into one of the two receivers. In still other aspects, a fastener is provided for each receiver configured to secure the protrusion to the corresponding receiver, wherein the backrest is removable upon removal of the fasteners. In yet other aspects the frame defines an opening which is a through opening and at least part of the seat is arranged to move into said opening when a user sits on the seat. In still other aspects the frame defines an opening which is a through opening and at least part of the seat is arranged to move through said opening when a user sits on the seat. In yet other aspects the opening is defined by at least four straight sections of

the frame.
(12) Other objects of the invention and its particular features and advantages will become more apparent from consideration of the following drawings and accompanying detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective view of a chair according to the present invention.
(2) FIG. **2** is a perspective view of the chair of FIG. **1** with certain components removed.
(3) FIG. **3** is a perspective view of a component of the chair of FIG. **1**.
(4) FIG. **4** is a perspective view of the bottom frame section of the chair of FIG. **1**.
(5) FIG. **5** is a side view of the backrest of the chair of FIG. **1**.
(6) FIG. **6** is a rear view of the backrest of FIG. **5**.
(7) FIG. **7** is a perspective view of the backrest of FIG. **5**.
(8) FIG. **8** is a detail view of part of FIG. **6**.
(9) FIG. **9** is a top view of the chair of FIG. **1** with the backrest removed.
(10) FIG. **10** shows a perspective view of multiple chairs according to FIG. **1** arranged in rows.
(11) FIG. **11** is a rear view of the chair of FIG. **1**.
(12) FIG. **12** is a detail view of FIG. **13**.
(13) FIG. **13** is a rear view of the chair of FIG. **1** with the backrest removed.
(14) FIG. **14** is a bottom perspective view the chair of FIG. **1**.
(15) FIG. **15** is a perspective view of multiple chairs according to FIG. **1** stacked on top of each other.
DETAILED DESCRIPTION OF THE INVENTION
(16) Referring now to the drawings, wherein like reference numerals designate corresponding structure throughout the views. The following examples are presented to further illustrate and explain the present invention and should not be taken as limiting in any regard.
(17) The chair **1** includes a backrest **2** and a seat **4**. The seat **4** sits on top of frame **6** and the seat **4** is in a fixed position relative to the frame as is the backrest in that the seat **4** and backrest **2** do not have a tilting function. Connector piece **11** can be provided to assist in holding chairs together/in place in rows (FIG. **10**). As can be seen in FIG. **2**, the seat **4** can be removed. The backrest **2** and seat **4** are upholstered, thus the removal of the seat assists in replacing the upholstery cover in the even the originally installed one becomes worn, damaged or dirty. The seat **4** includes a support structure shown in FIG. **3** which is positioned under the foam cushion and on top of the lower frame (FIG. **4**). The cushion support frame pieces Dec. 14, 2016/18 are preferably welded together and a flexible support **20** is placed over/around these pieces. On the seat frame, piece **14** is curved to match the curve of the corresponding frame section **62**. The seat frame pieces Dec. 14, 2016/18 are preferably made from a relatively thin material, for example ½″ or thinner metal, more preferably ⅜″ or ¼″ or thinner metal such as steel. In preferred embodiments, the metal includes threaded holes to receive the screws that secure the seat **4** to the frame **6** from underneath.
(18) Due to the use of the flexible support **20** and the foam placed thereon or molded there around, the bottom of the seat **4** will displace into or through the opening **7** when the seat is weighted, for example when someone sits on the seat. This flexible support **20** may be webbing or a seat sling material as example, but other materials may be used. A more rigid material may also be used for the support **20**. A piece of foam rests on top of the flexible support **20** and the upholstery cover is placed over that assembly. The support frame can be attached/removed from the lower frame (FIG. **4**) with screws. In this way, the screws can be removed and the seat frame and cushion pulled off the chair. The upholstery cover shown on the seat **4** is preferably removable, for example with a zipper, hook and loop fasteners (e.g. Velcro) or the like using a flap whose seam is shown in FIG.

11 (**31**). This allows for replacement/cleaning/repair of the upholstery cover as needed. Often chairs of the design shown are used in places of worship and thus the ability for user replacement of the upholstery is an important value feature. Specifically, places of worship would be able to gather volunteers to do a project of replace/renew all of the upholstery on the stacking chairs. As a result, the labor can be donated by members and the cost of having near new chairs is only the cost of new upholstery covers. Thus, the ease of re-upholstery for fixed seat stacking chairs is a significant advantage. Additionally, commercial venues can also replace upholstery with relative ease and a modest labor expense.
(19) FIG. **4** shows additional details on the lower frame construction. Leg **60** is connected to curved section **62** which is connected to horizontal section **64**. Another curved section is connected between horizontal section **64** and the rear leg **68**. All of these sections **60**/**62**/**64**/**68** of the frame are fixed relative to each other, e.g. by welding or the sections can be formed by bending square tubing. Cross bars **10**/**8** connect each side of the frame and cross bar **10** includes back supports **22**, preferably a welded connection is used. Preferably the frame pieces are hollow and particularly the back support **22** is hollow. Generally the square tubing will have chamfered or rounded corners. As shown in FIGS. **5-8**, the back includes lower extensions **24** or protrusions which protrude from below the backrest's upholstered cushion. Like the seat upholstery, the back upholstery also has removable fasteners to enable removal and replacement (or cleaning) of the upholstery cover. FIG. **14** shows the flap **33** which can be opened to allow for removal of the upholstery cover. The lower extension **24** is shown as a round tube and boss **26** provides a flat surface to interface with the interior square tube of the back supports **22**. The box **26** also includes a threaded hole, allowing screws to secure the backrest to the lower frame. Further, the lower section of the back **35** is narrower than the upper section **37** and there are curved and/or angled sections between the narrower **35** and wider **37** sections.
(20) As shown in FIGS. **9**, **11** and **12**, the back supports **22** are hollow and include a hole **28** through which a screw **30** can pass to secure the backrest **2** to the lower frame **6**. These screws can also be removed to allow the backrest to likewise be removed. The preferred configuration when shipping chairs is with the backrest removed. Shipping typically has constrained height spaces in that the stack of chairs will cause interference with the backrest and the ceiling of the container/truck in which the chairs are placed. This space **140** is not well used in that the height of the backrest would allow about two or three additional chairs to be accommodated in that otherwise empty vertical space. This can be seen in the stack at FIG. **15** in that the top of the back of the lowest chair is roughly at the level of the seat of the 4.sup.th chair in the stack **142**, certainly two lower frames with upholstery and no backrest can fit in this space **140**. Then, the backrests can be nested together in a stack with maximum density in that the front surface of one back is in contact with a rear surface of another and so on. Thus, there is minimum empty space in the container/truck, and specifically as to the backrests. As a result, more chairs can be shipped within a given space constraint.
(21) Although the invention has been described with reference to a particular arrangement of parts, features and the like, these are not intended to exhaust all possible arrangements or features, and indeed many other modifications and variations will be ascertainable to those of skill in the art.

Claims

1. A stacking chair comprising: a frame with four legs, two being front legs and two being rear legs; a seat secured to the frame in a fixed position, the seat being upholstered and including a foam cushion supported by flexible support; two receivers in the frame at a rear of the frame; a backrest comprising a foam cushion which is upholstered and an inner frame supporting the foam cushion, the inner frame including two protrusions extending out of the upholstery wherein the each one of the two protrusions is configured to insert into one of the two receivers; a fastener for each receiver

configured to secure the protrusion to the corresponding receiver, wherein the backrest is removable upon removal of the fasteners.

2. The stacking chair of claim 1 wherein the frame further comprises a support bar extending on either side of the seat and connected to a front curved section and a rear curved section, the front curved section connected to one of the front legs and the rear curved section connected to one of the rear legs, each of the four legs extending outwards and downwards with respect to the seat and the four legs in a fixed arrangement with respect to the seat.

3. The stacking chair of claim 2 wherein the two front legs are arranged at a more vertical angle than the rear legs.

4. The stacking chair of claim 2 wherein the seat includes rigid supports arranged around a peripheral edge thereof to define a rigid frame and wherein the flexible support is secured to the rigid frame and the rigid frame is removably secured to a side section of the frame.

5. The stacking chair of claim 4 wherein the foam cushion of the seat is arranged on top of the rigid frame.

6. The stacking chair of claim 1 wherein the frame further comprises a rear cross bar and the each of two receivers are arranged as a tube extending from the cross bar, the two receivers spaced apart at a distance which is at least 50% or more a width of the seat, but less than a spacing between the rear legs.

7. The stacking chair of claim 1 wherein the seat includes a seat frame and the flexible support is secured to the seat frame, the seat frame includes a curved portion such that the seat includes a lower curved section which matches a curve of the frame adjacent the front legs.

8. The stacking chair of claim 7 wherein the seat frame is secured to the frame such that the seat frame rests on top of the frame with upholstery between the seat and the seat frame and fasteners passing through the frame and securing to the seat frame.

9. The stacking chair of claim 1 wherein the receiver defines an opening with four flat sides.

10. The stacking chair of claim 9 wherein the protrusion includes a circular cross section.

11. The stacking chair of claim 10 wherein the protrusion includes a boss defining a flat section and having a threaded feature adjacent thereto for receiving the fasteners.

12. The stacking chair of claim 1 wherein the frame comprises a rear cross bar and the receivers are secured to the rear cross bar at an attachment point which is lower than a lower surface of a flat section of the frame which is perpendicular to the rear cross bar.

13. A stacking chair comprising: a frame with four legs, two being front legs and two being rear legs; a seat secured to the frame in a fixed position, the seat including a seat frame which defines a perimeter with an interior opening and the seat frame includes at least a flat portion and a curved portion, the curved portion having a curve which matches a curve of the frame at or adjacent the front legs and the flat portion configured to rest on top of flat portions of the frame, the seat frame contained in upholstery and the seat including a foam cushion positioned above the seat frame wherein the foam cushion is supported by a flexible support, the flexible support connected to the seat frame and extending across the interior opening.

14. The stacking chair of claim 13 further comprising seat fasteners which pass through holes in the flat portions of the frame and through the upholstery to secure to threaded members of the seat frame.

15. The stacking chair of claim 13 wherein at least part of the upholstery is positioned between the seat frame and the flat portions of the frame with the upholstery in contact with the flat portions of the frame.

16. The stacking chair of claim 13 further comprising: two receivers in the frame at a rear of the frame and between the rear legs of the frame; a backrest comprising a foam cushion which is upholstered and an inner frame supporting the foam cushion, the inner frame including two protrusions extending out of the upholstery wherein the each one of the two protrusions is configured to insert into one of the two receivers; a fastener for each receiver configured to secure

the protrusion to the corresponding receiver, wherein the backrest is removable upon removal of the fasteners.
17. The stacking chair of claim 13 wherein the frame defines an opening which is a through opening and at least part of the seat is arranged to move into said opening when a user sits on the seat.
18. The stacking chair of claim 17 wherein the opening is defined by at least four straight sections of the frame.